

Unit 1 :: Partial Differential Equations

☞ **PHY0400404: Mathematical Physics** ☞

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Syllabus

Unit 1: Partial differential equations

Solutions to partial differential equations, using separation of variables: Laplace's equation in problems of rectangular, cylindrical and spherical symmetry. Wave equation and its solution for vibrational modes of a stretched string, rectangular and circular membranes.

1 Introduction to partial differential equations

A partial differential equation (or PDE) is an equation that involves an unknown function of two or more variables and its partial derivatives.

For an unknown function $u(x_1, x_2, \dots, x_n)$, a general form of a PDE is given below:

$$F\left(x_1, x_2, \dots, x_n, u, \frac{\partial u}{\partial x_i}, \frac{\partial^2 u}{\partial x_i \partial x_j}, \dots\right) = 0$$

1.1 Classification of PDEs

There are mainly four ways to classify PDEs: **Homogeneity**, **Order**, **Linearity** and **Degree** (remember: *HOLD*).

1.1.1 Order

The *order* of a PDE is the order of the highest partial derivative.

Examples:

PDE	Highest derivative	Order
$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$	first	1
$\frac{\partial u}{\partial t} - u \frac{\partial u}{\partial x} = 0$	first	1
$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$	second	2
$\frac{\partial^2 u}{\partial t^2} - \alpha \frac{\partial u}{\partial x} = 0$	second	2
$\frac{\partial^3 u}{\partial x^3} - \frac{\partial u}{\partial y} = 0$	third	3

1.1.2 Degree

The *degree* of a PDE is the power of the highest derivative when the PDE is a polynomial in derivatives.

Examples:

PDE	Highest derivative	Order	Degree
$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$	first	1	1
$\frac{\partial^2 u}{\partial x^2} + \left(\frac{\partial u}{\partial t}\right)^2 = 0$	second	2	1
$\left(\frac{\partial u}{\partial x}\right)^2 + \frac{\partial u}{\partial y} = 5$	first	1	2
$\frac{\partial^2 u}{\partial x^2} - \sqrt{1 + \left(\frac{\partial u}{\partial y}\right)^2} = 0$	second	2	2

1.1.3 Linearity

A PDE is said to be *linear* if the unknown function and its derivatives appear linearly, i.e. there are no products of its derivatives or no non-linear functions of derivatives; otherwise the PDE is said to be *non-linear*.

Examples:

PDE	Linearity
$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$	linear
$\frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} = 0$	linear
$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	linear
$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$	non-linear
$\frac{\partial u}{\partial t} = \left(\frac{\partial u}{\partial x}\right)^2$	non-linear
$\left(\frac{\partial^2 u}{\partial x^2}\right)^2 + \frac{\partial^2 u}{\partial y^2} = 0$	non-linear

1.1.4 Homogeneity

A PDE is said to be *homogeneous* if every term in the equation involves the unknown function or one of its derivatives, and no terms that do not depend on the unknown function; otherwise the PDE is said to be *inhomogeneous*.

Examples:

PDE	Homogeneity
$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$	homogeneous
$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	homogeneous
$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$	homogeneous
$\frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} = x e^{-t^2}$	inhomogeneous
$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 5 \sinh x^2 - 3 \cosh t$	inhomogeneous
$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = x$	inhomogeneous

1.2 Second order, first degree linear PDEs

Second order, first degree linear PDEs are of particular interest in Physics because they appear in many different scenarios. Also they are easier to handle because their solutions are superposable as long as the equation is linear and homogenous.

Such equations have the following general form:

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial t} + C \frac{\partial^2 u}{\partial t^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial t} + F u = G$$

where, in general, the coefficients A, B, C, D, E, F and G may depend on x and t .

Such PDEs may be further classified as follows:

$$B^2 - 4AC \begin{cases} < 0 & \Rightarrow \text{elliptic} \\ = 0 & \Rightarrow \text{parabolic} \\ > 0 & \Rightarrow \text{hyperbolic} \end{cases}$$

For instance, consider the Laplace equation in two dimensions:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Considering $y \rightarrow t$ and comparing with the general form, we find that

$$A = 1, B = 0, C = 1$$

which gives us

$$\begin{aligned} B^2 - 4AC &= 0^2 - 4 \cdot 1 \cdot 1 = -4 \\ \Rightarrow B^2 - 4AC &< 0 \end{aligned}$$

Therefore the *Laplace equation in two dimensions is elliptic*.

1.3 Standard PDEs in Physics

1. Euler equation:

$$\boxed{\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{\nabla P}{\rho}}$$

A first order, first degree, non-linear, inhomogeneous PDE.

2. Laplace equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

$$\text{or, } \boxed{\nabla^2 u = 0}$$

A second order, first degree, linear, homogeneous PDE (elliptic).

3. Poisson equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f(x, y, z)$$

$$\text{or, } \boxed{\nabla^2 u = f(x, y, z)}$$

A second order, first degree, linear, inhomogeneous PDE (elliptic).

4. Helmholtz equation:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \pm k^2 \psi = 0$$

$$\text{or, } \boxed{\nabla^2 \psi \pm k^2 \psi = 0}$$

A second order, first degree, linear, homogeneous PDE (elliptic).

5. Diffusion equation:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{a^2} \frac{\partial \psi}{\partial t}$$

$$\text{or, } \boxed{\nabla^2 \psi = \frac{1}{a^2} \frac{\partial \psi}{\partial t}}$$

A second order, first degree, linear, homogeneous PDE (parabolic).

6. Wave equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$\text{or, } \boxed{\nabla^2 u = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}}$$

A second order, first degree, linear, homogeneous PDE (hyperbolic).

2 Separation of variables

The method of *separation of variables* is a procedure to solve some linear and homogeneous PDEs by assuming that the solution can be written as a product of functions, with each function depending on only one independent variable. Substituting this product form into the PDE allows the equation to be separated into simpler ordinary differential equations (ODEs).

Suppose we want to solve a PDE for an unknown function u of two variables x and t , i.e. $u = u(x, t)$. We assume that

$$u(x, t) = X(x)T(t)$$

where the function X depends only on x and the function T depends only on t .

When this form is substituted into the original PDE, the result often allows us to separate the variables so that all terms involving x appear on one side and all terms involving t on the other. Because the two sides depend on different independent variables, each side must equal a constant. This leads to a pair of ODEs — one for each variable.

- The method is typically used for linear, homogeneous PDEs with appropriate boundary conditions.
- The method is especially useful when the domain has simple geometry (e.g. rectangular or cylindrical) and when one can choose coordinates that allow the PDE to be separated.
- Common equations solved this way include the heat equation, wave equation and Laplace's equation.

2.1 Basic procedure

- I. We assume a separable solution of the form

$$u(x, t) = X(x)T(t)$$

- II. We substitute $X(x)T(t)$ for $u(x, t)$ in the PDE and simplify.
- III. The equation is divided throughout by $X(x)T(t)$ to isolate the terms in variable x on one side and the terms in variable t on the other side.
- IV. Because either side depends on variable that are independent of each other, it follows that both expressions must equal a constant. This leads to two ODEs.
- V. The ODEs are solved to obtain $X(x)$ and $T(t)$, which are then combined to obtain the general solution $u(x, t)$.
- VI. Any given initial or boundary conditions are applied on $u(x, t)$ to obtain the complete solution.

2.2 Examples

1. Find the solution of the following PDE using the method of separation of variables:

$$2\frac{\partial u}{\partial x} + 5\frac{\partial u}{\partial y} = 0.$$

Solution: We assume the solution

$$u(x, y) = X(x)Y(y) \quad (1)$$

This gives us

$$\frac{\partial u}{\partial x} = Y(y)\frac{dX}{dx} \quad \text{and} \quad \frac{\partial u}{\partial y} = X(x)\frac{dY}{dy}$$

Substituting in the given PDE, we obtain

$$\begin{aligned} 2Y(y)\frac{dX}{dx} + 5X(x)\frac{dY}{dy} &= 0 \\ \Rightarrow 2Y(y)\frac{dX}{dx} &= -5X(x)\frac{dY}{dy} \end{aligned} \quad (2)$$

Dividing equation (2) by $X(x)Y(y)$, we get

$$\begin{aligned} \frac{1}{X(x)Y(y)} \times 2Y(y)\frac{dX}{dx} &= -5X(x)\frac{dY}{dy} \times \frac{1}{X(x)Y(y)} \\ \Rightarrow \frac{2}{X(x)}\frac{dX}{dx} &= -\frac{5}{Y(y)}\frac{dY}{dy} \end{aligned} \quad (3)$$

In equation (3), the LHS depends only on x and the RHS depends only on y . Because x and y are independent, the only way for equation (3) to hold good is if both sides are simultaneously equal to the same constant k , i.e.

$$\begin{aligned} \frac{2}{X(x)}\frac{dX}{dx} &= k = -\frac{5}{Y(y)}\frac{dY}{dy} \\ \Rightarrow \frac{2}{X(x)}\frac{dX}{dx} &= k \end{aligned} \quad (4)$$

$$\text{and} \quad \frac{5}{Y(y)}\frac{dY}{dy} = -k \quad (5)$$

To solve equation (4), we have

$$\begin{aligned} \frac{2}{X(x)}\frac{dX}{dx} &= k \Rightarrow \frac{dX}{X} = \frac{k}{2}dx \\ \Rightarrow \int \frac{dX}{X} &= \int \frac{k}{2}dx \\ \Rightarrow \ln X &= \frac{kx}{2} + \ln c_1 \quad [\text{where } \ln c_1 \text{ is the integration constant}] \\ \Rightarrow \ln X - \ln c_1 &= \frac{kx}{2} \\ \Rightarrow \ln\left(\frac{X}{c_1}\right) &= \frac{kx}{2} \\ \Rightarrow \frac{X}{c_1} &= e^{kx/2} \\ \Rightarrow X(x) &= c_1 e^{kx/2} \end{aligned} \quad (6)$$

Similarly, to solve equation (5), we have

$$\begin{aligned}
 \frac{5}{Y(y)} \frac{dY}{dy} &= -k \implies \frac{dY}{Y} = -\frac{k}{5} dy \\
 \implies \int \frac{dY}{Y} &= -\int \frac{k}{5} dy \\
 \implies \ln Y &= -\frac{ky}{5} + \ln c_2 \quad [\text{where } \ln c_2 \text{ is the integration constant}] \\
 \implies \ln Y - \ln c_2 &= -\frac{ky}{5} \\
 \implies \ln \left(\frac{Y}{c_2} \right) &= -\frac{ky}{5} \\
 \implies \frac{Y}{c_2} &= e^{-ky/5} \\
 \implies Y(y) &= c_2 e^{-ky/5}
 \end{aligned} \tag{7}$$

Finally, substituting equations (6) and (7) into equation (1)

$$\begin{aligned}
 u(x, y) &= X(x)Y(y) \\
 &= (c_1 e^{kx/2})(c_2 e^{-ky/5}) \\
 &= \underbrace{c_1 c_2}_{=c} e^{k(x/2 - y/5)} \\
 \implies \boxed{u(x, y) &= c e^{k(x/2 - y/5)}}
 \end{aligned}$$

which gives us the general solution of the PDE.

2. Find the solution of the following PDE using the method of separation of variables:

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}, \text{ with } u(0, y) = 8e^{-3y}.$$

Solution: We assume the solution

$$u(x, y) = X(x)Y(y) \tag{1}$$

This gives us

$$\frac{\partial u}{\partial x} = Y(y) \frac{dX}{dx} \quad \text{and} \quad \frac{\partial u}{\partial y} = X(x) \frac{dY}{dy}$$

Substituting in the given PDE, we obtain

$$Y(y) \frac{dX}{dx} = 4X(x) \frac{dY}{dy} \tag{2}$$

Dividing equation (2) by $X(x)Y(y)$, we get

$$\begin{aligned}
 \frac{1}{\cancel{X(x)} \cancel{Y(y)}} \times \cancel{Y(y)} \frac{dX}{dx} &= 4 \cancel{X(x)} \frac{dY}{dy} \times \frac{1}{\cancel{X(x)} \cancel{Y(y)}} \\
 \implies \frac{1}{X(x)} \frac{dX}{dx} &= \frac{4}{Y(y)} \frac{dY}{dy}
 \end{aligned} \tag{3}$$

In equation (3), the LHS depends only on x and the RHS depends only on y . Because x and y are independent, the only way for equation (3) to hold good is if both sides are simultaneously equation to the same constant k , i.e.

$$\begin{aligned} \frac{1}{X(x)} \frac{dX}{dx} &= k = \frac{4}{Y(y)} \frac{dY}{dy} \\ \Rightarrow \frac{1}{X(x)} \frac{dX}{dx} &= k \end{aligned} \quad (4)$$

$$\text{and } \frac{4}{Y(y)} \frac{dY}{dy} = k \quad (5)$$

To solve equation (4), we have

$$\begin{aligned} \frac{1}{X(x)} \frac{dX}{dx} &= k \Rightarrow \frac{dX}{X} = k dx \\ \Rightarrow \int \frac{dX}{X} &= \int k dx \\ \Rightarrow \ln X &= kx + \ln c_1 \quad [\text{where } \ln c_1 \text{ is the integration constant}] \\ \Rightarrow \ln X - \ln c_1 &= kx \\ \Rightarrow \ln \left(\frac{X}{c_1} \right) &= kx \\ \Rightarrow \frac{X}{c_1} &= e^{kx} \\ \Rightarrow X(x) &= c_1 e^{kx} \end{aligned} \quad (6)$$

Similarly, to solve equation (5), we have

$$\begin{aligned} \frac{4}{Y(y)} \frac{dY}{dy} &= k \Rightarrow \frac{dY}{Y} = \frac{k}{4} dy \\ \Rightarrow \int \frac{dY}{Y} &= \int \frac{k}{4} dy \\ \Rightarrow \ln Y &= \frac{ky}{4} + \ln c_2 \quad [\text{where } \ln c_2 \text{ is the integration constant}] \\ \Rightarrow \ln Y - \ln c_2 &= \frac{ky}{4} \\ \Rightarrow \ln \left(\frac{Y}{c_2} \right) &= \frac{ky}{4} \\ \Rightarrow \frac{Y}{c_2} &= e^{ky/4} \\ \Rightarrow Y(y) &= c_2 e^{ky/4} \end{aligned} \quad (7)$$

Finally, substituting equations (6) and (7) into equation (1)

$$\begin{aligned} u(x, y) &= X(x)Y(y) \\ &= (c_1 e^{kx})(c_2 e^{ky/4}) \\ &= \underbrace{c_1 c_2}_{=c} e^{k(x+y/4)} \\ \Rightarrow u(x, y) &= c e^{k(x+y/4)} \end{aligned} \quad (8)$$

which gives us the general solution of the PDE.

It is also given that $u(0, y) = 8e^{-3y}$. With $x = 0$ in equation (8), we obtain

$$\begin{aligned} u(0, y) &= c e^{k(0+y/4)} \\ \implies 8e^{-3y} &= c e^{ky/4} \end{aligned}$$

Comparing both sides in the above, we find that

$$c = 8 \quad \text{and} \quad \frac{k}{4} = -3 \implies k = -12$$

Substituting for c, k in equation (8) gives us the complete solution as

$$u(x, y) = 8e^{-12(x+y/4)}$$

Exercises

- Find the general solution of the following PDE using the method of separation of variables:

$$y^3 \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} = 0.$$

- Solve the following PDE using the method of separation of variables:

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2(x + y)u.$$

- Find the complete solution of the following PDE using the method of separation of variables:

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial y} + u, \text{ with } u(x, 0) = 4e^{-3x}.$$

- Solve the following PDE using the method of separation of variables:

$$\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0.$$

- Find the complete solution of the following PDE using the method of separation of variables:

$$\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$$

given that $u = 0$ when $t = 0$, and $\frac{\partial u}{\partial t} = 0$ when $x = 0$.

- Solve the following PDEs using the method of separation of variables:

$$(a) \quad \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial y} = 0$$

$$(b) \quad 4 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u \text{ with } u(0, y) = e^{-5y}$$

$$(c) \quad 2x \frac{\partial z}{\partial x} - 3y \frac{\partial z}{\partial y} = 0$$

$$(d) \quad 4 \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 3u \text{ with } u(x, 0) = 3e^{-x} - e^{-5x}$$

3 Laplace's equation

The Laplace equation is a second-order, linear, homogeneous PDE. It is expressed as follows:

- In Cartesian coordinates (x, y, z) :

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

- In cylindrical coordinates (s, ϕ, z) :

$$\nabla^2 u = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial u}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

- *Special case:* In case of azimuthal symmetry, we have $\frac{\partial u}{\partial \phi} = 0$, and so the Laplace equation simplifies to

$$\frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial u}{\partial s} \right) + \frac{\partial^2 u}{\partial z^2} = 0$$

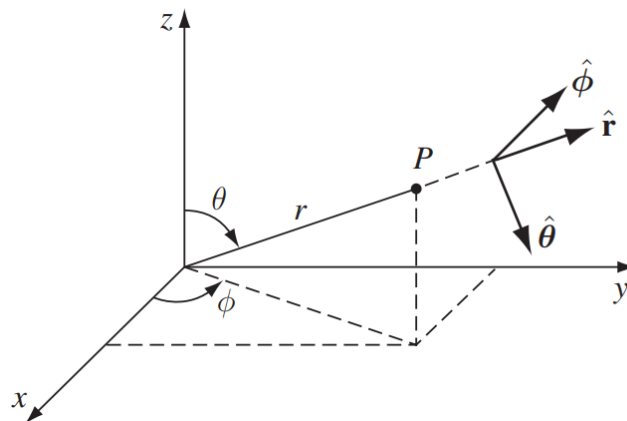
- In spherical coordinates (r, θ, ϕ) :

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} = 0$$

- *Special case:* In case of azimuthal symmetry, we have $\frac{\partial u}{\partial \phi} = 0$, and so the Laplace equation simplifies to

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = 0$$

Review: spherical coordinate system



In a spherical coordinate system, any arbitrary point P can be labelled by the spherical coordinates (r, θ, ϕ) , where the *radial distance* r , *polar angle* θ and *azimuthal angle* ϕ are related to the Cartesian coordinates (x, y, z) as

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

At any position, the set of orthogonal unit vectors, $\hat{\mathbf{r}}$, $\hat{\theta}$ and $\hat{\phi}$, point in the direction of increase of the corresponding coordinates. These unit vectors are related to the Cartesian unit vectors as follows:

$$\begin{aligned}\hat{\mathbf{r}} &= \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}} \\ \hat{\theta} &= \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}} \\ \hat{\phi} &= -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}\end{aligned}$$

For a given vector $\mathbf{A}$ if A_r , A_θ and A_ϕ represent its radial, polar and azimuthal components, then we have

$$\mathbf{A} = A_r \hat{\mathbf{r}} + A_\theta \hat{\theta} + A_\phi \hat{\phi}$$

The vector derivatives in spherical coordinates are as follows:

- *Gradient*: For a given scalar field u , we have

$$\nabla u = \frac{\partial u}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial u}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi} \hat{\phi}$$

- *Divergence*: For a given vector field $\mathbf{A}$, we have

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

- *Curl*: For a given vector field $\mathbf{A}$, we have

$$\begin{aligned}\nabla \times \mathbf{A} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \hat{\theta} \\ &\quad + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\phi}\end{aligned}$$

3.1 Laplace's equation in spherical coordinates

For any scalar field u , its Laplacian is obtained as the divergence of its curl, i.e.

$$\nabla^2 u = \nabla \cdot (\nabla u) \quad (1)$$

We know that the gradient of u in spherical coordinates is

$$\nabla u = \frac{\partial u}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial u}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi} \hat{\phi} \quad (2)$$

Let $\mathbf{A}$ be a vector field, such that $\mathbf{A} = \nabla u$. In terms of its radial, polar and azimuthal components, we can write

$$\begin{aligned}\mathbf{A} &= A_r \hat{\mathbf{r}} + A_\theta \hat{\theta} + A_\phi \hat{\phi} \\ \Rightarrow \nabla u &= A_r \hat{\mathbf{r}} + A_\theta \hat{\theta} + A_\phi \hat{\phi}\end{aligned} \quad (3)$$

Comparing equations (2) and (3), we obtain

$$A_r = \frac{\partial u}{\partial r}, \quad A_\theta = \frac{1}{r} \frac{\partial u}{\partial \theta}, \quad A_\phi = \frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi}$$

Now, we know that, the divergence of the vector field $\mathbf{A}$ in spherical coordinates is

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

Substituting for A_r , A_θ and A_ϕ above, we obtain

$$\begin{aligned} \nabla \cdot (\nabla u) &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{1}{r} \frac{\partial u}{\partial \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi} \right) \\ \Rightarrow \nabla^2 u &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} \end{aligned} \quad (4)$$

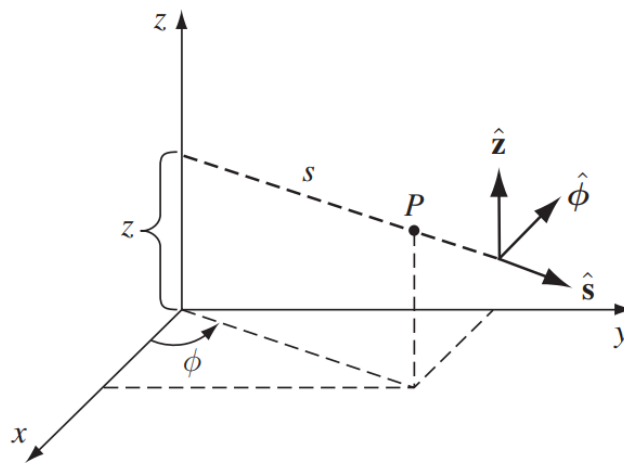
Using equation (4), we finally obtain the Laplace's equation in spherical coordinates as

$$\begin{aligned} \nabla^2 u &= 0 \\ \Rightarrow \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} &= 0 \end{aligned}$$

An alternative form of the Laplace's equation in spherical coordinates is obtained by applying the product rule to the first term. This form is more convenient in certain boundary value problems

$$\frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} = 0$$

Review: cylindrical coordinate system



In a cylindrical coordinate system, any arbitrary point P can be labelled by the cylindrical coordinates (s, ϕ, z) , where the *axial distance* s , *azimuthal angle* ϕ and *height* z are related to the Cartesian coordinates (x, y, z) as

$$x = s \cos \phi, \quad y = s \sin \phi, \quad z = z$$

At any position, the set of orthogonal unit vectors, $\hat{\mathbf{s}}$, $\hat{\phi}$ and $\hat{\mathbf{z}}$, point in the direction of increase of the corresponding coordinates. These unit vectors are related to the Cartesian unit vectors as follows:

$$\begin{aligned}\hat{\mathbf{s}} &= \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}} \\ \hat{\phi} &= -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}} \\ \hat{\mathbf{z}} &= \hat{\mathbf{z}}\end{aligned}$$

For a given vector $\mathbf{A}$ if A_s , A_ϕ and A_z represent its axial, azimuthal and height components, then we have

$$\mathbf{A} = A_s \hat{\mathbf{s}} + A_\phi \hat{\phi} + A_z \hat{\mathbf{z}}$$

The vector derivatives in cylindrical coordinates are as follows:

- *Gradient*: For a given scalar field u , we have

$$\nabla u = \frac{\partial u}{\partial s} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial u}{\partial \phi} \hat{\phi} + \frac{\partial u}{\partial z} \hat{\mathbf{z}}$$

- *Divergence*: For a given vector field $\mathbf{A}$, we have

$$\nabla \cdot \mathbf{A} = \frac{1}{s} \frac{\partial}{\partial s} (s A_s) + \frac{1}{s} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

- *Curl*: For a given vector field $\mathbf{A}$, we have

$$\nabla \times \mathbf{A} = \left(\frac{1}{s} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \hat{\mathbf{s}} + \left(\frac{\partial A_s}{\partial z} - \frac{\partial A_z}{\partial s} \right) \hat{\phi} + \frac{1}{s} \left[\frac{\partial}{\partial s} (s A_\phi) - \frac{\partial A_s}{\partial \phi} \right] \hat{\mathbf{z}}$$

3.2 Laplace's equation in cylindrical coordinates

For any scalar field u , its Laplacian is obtained as the divergence of its curl, i.e.

$$\nabla^2 u = \nabla \cdot (\nabla u) \quad (1)$$

We know that the gradient of u in cylindrical coordinates is

$$\nabla u = \frac{\partial u}{\partial s} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial u}{\partial \phi} \hat{\phi} + \frac{\partial u}{\partial z} \hat{\mathbf{z}} \quad (2)$$

Let $\mathbf{A}$ be a vector field, such that $\mathbf{A} = \nabla u$. In terms of its axial, azimuthal and height components, we can write

$$\begin{aligned}\mathbf{A} &= A_s \hat{\mathbf{s}} + A_\phi \hat{\phi} + A_z \hat{\mathbf{z}} \\ \Rightarrow \nabla u &= A_s \hat{\mathbf{s}} + A_\phi \hat{\phi} + A_z \hat{\mathbf{z}}\end{aligned} \quad (3)$$

Comparing equations (2) and (3), we obtain

$$A_s = \frac{\partial u}{\partial s}, \quad A_\phi = \frac{1}{s} \frac{\partial u}{\partial \phi}, \quad A_z = \frac{\partial u}{\partial z}$$

Now, we know that, the divergence of the vector field $\mathbf{A}$ in cylindrical coordinates is

$$\nabla \cdot \mathbf{A} = \frac{1}{s} \frac{\partial}{\partial s}(s A_s) + \frac{1}{s} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

Substituting for A_s , A_ϕ and A_z above, we obtain

$$\begin{aligned} \nabla \cdot (\nabla u) &= \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial u}{\partial s} \right) + \frac{1}{s} \frac{\partial}{\partial \phi} \left(\frac{1}{s} \frac{\partial u}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial z} \right) \\ \Rightarrow \nabla^2 u &= \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial u}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} \end{aligned} \quad (4)$$

Using equation (4), we finally obtain the Laplace's equation in cylindrical coordinates as

$$\begin{aligned} \nabla^2 u &= 0 \\ \Rightarrow \boxed{\frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial u}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} = 0} \end{aligned}$$

An alternative form of the Laplace's equation in cylindrical coordinates is obtained by applying the product rule to the first term. This form is more convenient in certain boundary value problems

$$\boxed{\frac{\partial^2 u}{\partial s^2} + \frac{1}{s} \frac{\partial u}{\partial s} + \frac{1}{s^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} = 0}$$

Note: It is helpful to remember the general forms of the following second order ODEs:

- The general solution of the ODE $\frac{d^2 f}{dx^2} - \lambda^2 f = 0$ is *exponential* in nature and given as

$$f(x) = c_1 e^{\lambda x} + c_2 e^{-\lambda x}$$

- The general solution of the ODE $\frac{d^2 f}{dx^2} + \lambda^2 f = 0$ is *harmonic* in nature and given as

$$f(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

Examples

1. Solve the Laplace equation over a rectangle in the xy -plane for the unknown function $u(x, y)$ with the following boundary conditions:

$$\begin{aligned} u(x, 0) &= 0, \quad u(x, b) = 0 \\ u(0, y) &= 0, \quad u(a, y) = f(y). \end{aligned}$$

Solution: The Laplace equation in the xy -plane is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (1)$$

We assume a solution of the form:

$$u(x, y) = X(x)Y(y) \quad (2)$$

which gives us

$$\frac{\partial^2 u}{\partial x^2} = Y \frac{d^2 X}{dx^2} \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = X \frac{d^2 Y}{dy^2}.$$

Then, substituting in equation (1), we have

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} = 0$$

Dividing the above throughout by $X(x)Y(y)$, we obtain

$$\begin{aligned} \frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} &= 0 \\ \Rightarrow \frac{1}{Y} \frac{d^2 Y}{dy^2} &= -\frac{1}{X} \frac{d^2 X}{dx^2} \end{aligned} \quad (3)$$

In equation (3), the LHS depends only on y , while the RHS depends only on x . Hence, the only way for this to be satisfied is if both side equal some separation constant.

Case I: the separation constant is positive, i.e.

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = -\frac{1}{X} \frac{d^2 X}{dx^2} = \lambda^2$$

We separate these into two ODEs as follows:

For the function $Y(y)$:

$$\begin{aligned} \frac{1}{Y} \frac{d^2 Y}{dy^2} &= \lambda^2 \\ \Rightarrow \frac{d^2 Y}{dy^2} &= \lambda^2 Y \\ \Rightarrow \frac{d^2 Y}{dy^2} - \lambda^2 Y &= 0 \end{aligned}$$

The above ODE has the general solution

$$Y(y) = c_1 e^{\lambda y} + c_2 e^{-\lambda y} \quad (4)$$

For the function $X(x)$:

$$\begin{aligned} -\frac{1}{X} \frac{d^2 X}{dx^2} &= \lambda^2 \\ \Rightarrow \frac{d^2 X}{dx^2} &= -\lambda^2 X \\ \Rightarrow \frac{d^2 X}{dx^2} + \lambda^2 X &= 0 \end{aligned}$$

The above ODE has the general solution

$$X(x) = c_3 \cos \lambda x + c_4 \sin \lambda x \quad (5)$$

From the boundary condition $u(x, 0) = 0$, we have that

$$\begin{aligned} X(x)Y(0) &= 0 \\ \Rightarrow Y(0) &= 0 \\ \Rightarrow c_1 e^0 + c_2 e^0 &= 0 \\ \Rightarrow c_2 &= -c_1 \end{aligned}$$

Again, from the boundary condition $u(x, b) = 0$, we have that

$$\begin{aligned}
 X(x)Y(b) &= 0 \\
 \implies Y(b) &= 0 \\
 \implies c_1 e^{\lambda b} + c_2 e^{-\lambda b} &= 0 \\
 \implies c_1 e^{\lambda b} - c_1 e^{-\lambda b} &= 0 \quad [\because c_2 = -c_1] \\
 \implies c_1 \underbrace{(e^{\lambda b} - e^{-\lambda b})}_{\neq 0} &= 0 \\
 \implies c_1 &= 0 \quad \text{and} \quad c_2 = 0
 \end{aligned}$$

This means that $Y(y) = 0$, thereby $u(x, y) = 0$ everywhere, which is not possible.

Therefore, this case with a positive separation constant is ruled out.

Case II: the separation constant is negative, i.e.

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = -\frac{1}{X} \frac{d^2 X}{dx^2} = -\lambda^2$$

We separate these into two ODEs as follows:

For the function $Y(y)$:

$$\begin{aligned}
 \frac{1}{Y} \frac{d^2 Y}{dy^2} &= -\lambda^2 \\
 \implies \frac{d^2 Y}{dy^2} &= -\lambda^2 Y \\
 \implies \frac{d^2 Y}{dy^2} + \lambda^2 Y &= 0
 \end{aligned}$$

The above ODE has the general solution

$$Y(y) = c_5 \cos \lambda y + c_6 \sin \lambda y \quad (6)$$

For the function $X(x)$:

$$\begin{aligned}
 -\frac{1}{X} \frac{d^2 X}{dx^2} &= -\lambda^2 \\
 \implies \frac{d^2 X}{dx^2} &= \lambda^2 X \\
 \implies \frac{d^2 X}{dx^2} - \lambda^2 X &= 0
 \end{aligned}$$

The above ODE has the general solution

$$X(x) = c_7 e^{\lambda x} + c_8 e^{-\lambda x} \quad (7)$$

Again, from the boundary condition $u(x, 0) = 0$, we have that

$$\begin{aligned}
 Y(0) &= 0 \\
 \implies c_5 \cos 0 + c_6 \sin 0 &= 0 \\
 \implies c_5 &= 0
 \end{aligned}$$

This gives us

$$Y(y) = c_6 \sin \lambda y \quad (8)$$

Applying the boundary condition $u(x, b) = 0$ on equation (8), we have

$$\begin{aligned}
 Y(b) &= c_6 \sin \lambda b = 0 \\
 \implies \sin \lambda b &= 0 \\
 \implies \sin \lambda b &= \sin n\pi \\
 \implies \lambda b &= n\pi \\
 \implies \lambda &= \frac{n\pi}{b}
 \end{aligned}$$

Then equations (7) and (8) become

$$X(x) = c_7 e^{n\pi x/b} + c_8 e^{-n\pi x/b} \quad (9)$$

$$Y(y) = c_6 \sin\left(\frac{n\pi y}{b}\right) \quad (10)$$

Inserting equations (9) and (10) into equation (2), we have

$$u(x, y) = c_6 \sin\left(\frac{n\pi y}{b}\right) [c_7 e^{n\pi x/b} + c_8 e^{-n\pi x/b}] \quad (11)$$

Applying the boundary condition $u(0, y) = 0$ on equation (11), we have

$$\begin{aligned} u(0, y) &= 0 \\ \Rightarrow \underbrace{c_6 \sin\left(\frac{n\pi y}{b}\right)}_{\neq 0} [c_7 e^0 + c_8 e^0] &= 0 \\ \Rightarrow c_7 + c_8 &= 0 \\ \Rightarrow c_8 &= -c_7 \end{aligned}$$

which gives us, from equation (11),

$$\begin{aligned} u(x, y) &= c_6 c_7 \sin\left(\frac{n\pi y}{b}\right) \underbrace{[e^{n\pi x/b} - e^{-n\pi x/b}]}_{=2\sinh(n\pi x/b)} \\ &= 2c_6 c_7 \sin\left(\frac{n\pi y}{b}\right) \sinh\left(\frac{n\pi x}{b}\right) \\ \Rightarrow u(x, y) &= C_n \sin\left(\frac{n\pi y}{b}\right) \sinh\left(\frac{n\pi x}{b}\right) \end{aligned} \quad (12)$$

where $C_n = 2c_6 c_7$.

Finally, applying the last boundary condition $u(a, y) = f(y)$ on equation (12), we have

$$\begin{aligned} u(a, y) &= f(y) \\ \Rightarrow C_n \sin\left(\frac{n\pi y}{b}\right) \sinh\left(\frac{n\pi a}{b}\right) &= f(y) \\ \Rightarrow C_n \sin\left(\frac{n\pi y}{b}\right) &= \frac{f(y)}{\sinh\left(\frac{n\pi a}{b}\right)} \\ \Rightarrow C_n \sin\left(\frac{n\pi y}{b}\right) \times \sin\left(\frac{n\pi y}{b}\right) &= \frac{f(y)}{\sinh\left(\frac{n\pi a}{b}\right)} \times \sin\left(\frac{n\pi y}{b}\right) \\ \Rightarrow C_n \sin^2\left(\frac{n\pi y}{b}\right) &= \frac{f(y)}{\sinh\left(\frac{n\pi a}{b}\right)} \sin\left(\frac{n\pi y}{b}\right) \\ \Rightarrow \underbrace{C_n \int_0^b \sin^2\left(\frac{n\pi y}{b}\right) dy}_{=b/2} &= \frac{1}{\sinh\left(\frac{n\pi a}{b}\right)} \int_0^b f(y) \sin\left(\frac{n\pi y}{b}\right) dy \\ \Rightarrow C_n \times \frac{b}{2} &= \frac{1}{\sinh\left(\frac{n\pi a}{b}\right)} \int_0^b f(y) \sin\left(\frac{n\pi y}{b}\right) dy \\ \Rightarrow C_n &= \frac{2}{b \sinh\left(\frac{n\pi a}{b}\right)} \int_0^b f(y) \sin\left(\frac{n\pi y}{b}\right) dy \end{aligned} \quad (13)$$

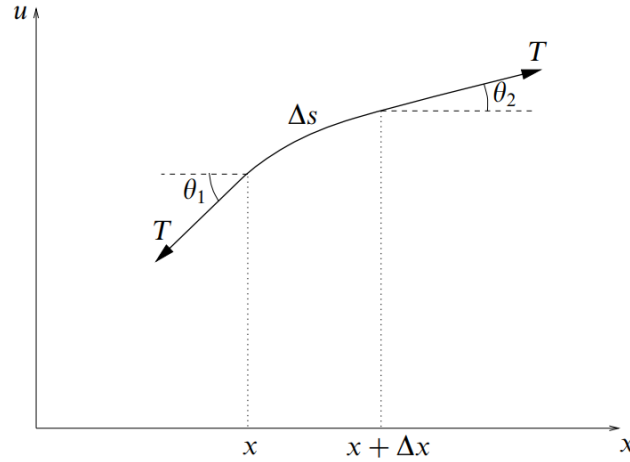
Therefore, the most general solution to the Laplace equation over the given rectangular region is given by

$$u(x, y) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi y}{b}\right) \sinh\left(\frac{n\pi x}{b}\right)$$

where the coefficients, given by equation (13), are known from the specified function $f(y)$.

4 Wave equation

Let us consider a pulse travelling on a string of linear mass density λ , with tension T .



The figure above shows the forces acting on a segment of length Δs on the string. The mass of the segment would be $\Delta m = \lambda \Delta s$. We find that the net upward force on this segment is

$$\begin{aligned}\Delta F &= T \sin \theta_2 - T \sin \theta_1 \\ \Rightarrow \Delta F &= T(\sin \theta_2 - \sin \theta_1)\end{aligned}\quad (1)$$

Assuming that the angles θ_1, θ_2 are small, we have the approximations:

$$\begin{aligned}\sin \theta_1 &\approx \tan \theta_1 = \frac{\partial}{\partial x} u(x, t) \\ \sin \theta_2 &\approx \tan \theta_2 = \frac{\partial}{\partial x} u(x + \Delta x, t)\end{aligned}$$

Now, given a function $f(x, t)$, its partial derivative is defined as

$$\frac{\partial f}{\partial x} = \frac{f(x + \Delta x, t) - f(x, t)}{\Delta x} \quad (2)$$

If we consider $f(x, t) = \frac{\partial u}{\partial x}$, then in equation (2), we have

$$\begin{aligned}\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) &= \frac{\frac{\partial}{\partial x} u(x + \Delta x, t) - \frac{\partial}{\partial x} u(x, t)}{\Delta x} \\ \Rightarrow \frac{\partial}{\partial x} u(x + \Delta x, t) - \frac{\partial}{\partial x} u(x, t) &= \frac{\partial^2 u}{\partial x^2} \Delta x \\ \Rightarrow \sin \theta_2 - \sin \theta_1 &= \frac{\partial^2 u}{\partial x^2} \Delta x\end{aligned}$$

Inserting the above in equation (1), we obtain

$$\Delta F = T \frac{\partial^2 u}{\partial x^2} \Delta x \quad (3)$$

The acceleration of the mass segment is given by $a = \frac{\partial^2 u}{\partial t^2}$. Hence, by Newton's second law of motion, we have

$$\begin{aligned} \Delta F &= (\Delta m)a \\ \Rightarrow T \frac{\partial^2 u}{\partial x^2} \Delta x &= (\lambda \Delta s) \left(\frac{\partial^2 u}{\partial t^2} \right) \quad [\text{using eqn. (3) on LHS}] \end{aligned} \quad (4)$$

If the vibrations on the string are small, we can approximate:

$$\Delta s \approx \Delta x$$

Therefore, in equation (4), we have

$$\begin{aligned} T \frac{\partial^2 u}{\partial x^2} \Delta x &= \lambda \Delta x \frac{\partial^2 u}{\partial t^2} \\ \Rightarrow \frac{\partial^2 u}{\partial x^2} &= \frac{\lambda}{T} \frac{\partial^2 u}{\partial t^2} \\ \Rightarrow \boxed{\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2}} \end{aligned} \quad (5)$$

Equation (5) is the *wave equation*, where $v = \sqrt{\frac{T}{\lambda}}$ is the speed of the wave.